In [1]:
```python
import pandas as pd
```

In [4]:
```python
pd.__version__
```

Out[4]: '2.2.3'

In [5]:
```python
data={"Milk Price List":[25,50,250],"Curd Price List":[15,30,350]}
```

In [6]:
```python
data
```

Out[6]: {'Milk Price List': [25, 50, 250], 'Curd Price List': [15, 30, 350]}

In [8]:
```python
df=pd.DataFrame(data)
df
```

Out[8]:

	Milk Price List	Curd Price List
0	25	15
1	50	30
2	250	350

In [31]:
```python
dataset_path="C:/Users/Student/Documents/DSML/dslab/iris.csv"
```

In [32]:
```python
df=pd.read_csv(dataset_path)
df
```

Out[32]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

150 rows × 5 columns

In [39]:
```python
cd "C:\Users\Student\Documents\DSML\dslab"
```

```
C:\Users\Student\Documents\DSML\dslab
```

In [41]:
```python
df=pd.read_csv('iris.csv')
df
```

Out[41]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

150 rows × 5 columns

In [42]:
```python
df.head()
```

Out[42]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

In [43]:
```python
df.tail()
```

Out[43]:

	sepal.length	sepal.width	petal.length	petal.width	variety
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

In [44]:
```python
df.head(20)
```

Out[44]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
5	5.4	3.9	1.7	0.4	Setosa
6	4.6	3.4	1.4	0.3	Setosa
7	5.0	3.4	1.5	0.2	Setosa
8	4.4	2.9	1.4	0.2	Setosa
9	4.9	3.1	1.5	0.1	Setosa
10	5.4	3.7	1.5	0.2	Setosa
11	4.8	3.4	1.6	0.2	Setosa
12	4.8	3.0	1.4	0.1	Setosa
13	4.3	3.0	1.1	0.1	Setosa
14	5.8	4.0	1.2	0.2	Setosa
15	5.7	4.4	1.5	0.4	Setosa
16	5.4	3.9	1.3	0.4	Setosa
17	5.1	3.5	1.4	0.3	Setosa
18	5.7	3.8	1.7	0.3	Setosa
19	5.1	3.8	1.5	0.3	Setosa

In [45]:
```python
df.tail(2)
```

Out[45]:

	sepal.length	sepal.width	petal.length	petal.width	variety
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

In [48]:
```python
df1=pd.read_csv('iris.csv',usecols=['sepal.length','variety'])
df1
```

Out[48]:

	sepal.length	variety
0	5.1	Setosa
1	4.9	Setosa
2	4.7	Setosa
3	4.6	Setosa
4	5.0	Setosa
...	...	...
145	6.7	Virginica
146	6.3	Virginica
147	6.5	Virginica
148	6.2	Virginica
149	5.9	Virginica

150 rows × 2 columns

In [50]:
```python
df2=df.set_index('variety')
df2
```

Out[50]:

	sepal.length	sepal.width	petal.length	petal.width
variety				
Setosa	5.1	3.5	1.4	0.2
Setosa	4.9	3.0	1.4	0.2
Setosa	4.7	3.2	1.3	0.2
Setosa	4.6	3.1	1.5	0.2
Setosa	5.0	3.6	1.4	0.2
...	...	...	...	...
Virginica	6.7	3.0	5.2	2.3
Virginica	6.3	2.5	5.0	1.9
Virginica	6.5	3.0	5.2	2.0
Virginica	6.2	3.4	5.4	2.3
Virginica	5.9	3.0	5.1	1.8

150 rows × 4 columns

In [51]:
```python
df2.to_csv('df_new.csv')
```

In [ ]:

DATASET INFORMATION

In [52]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [53]:
```python
df.describe()
```

Out[53]:

	sepal.length	sepal.width	petal.length	petal.width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [55]:
```python
df.shape
```

Out[55]: (150, 5)

In [56]:
```python
df.duplicated()
```

Out[56]:
```
0      False
1      False
2      False
3      False
4      False
       ...  
145    False
146    False
147    False
148    False
149    False
Length: 150, dtype: bool
```

In [57]:
```python
df.duplicated().sum()
```

Out[57]: np.int64(1)

In [59]:
```python
df.duplicated(subset='variety').sum()
```

Out[59]: np.int64(147)

In [60]:
```python
df=df.drop_duplicates()
```

In [61]:
```python
df
```

Out[61]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

149 rows × 5 columns

DATA SLICYING

In [64]:
```python
display(df.loc[(df.variety == 'Setosa')])
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
5	5.4	3.9	1.7	0.4	Setosa
6	4.6	3.4	1.4	0.3	Setosa
7	5.0	3.4	1.5	0.2	Setosa
8	4.4	2.9	1.4	0.2	Setosa
9	4.9	3.1	1.5	0.1	Setosa
10	5.4	3.7	1.5	0.2	Setosa
11	4.8	3.4	1.6	0.2	Setosa
12	4.8	3.0	1.4	0.1	Setosa
13	4.3	3.0	1.1	0.1	Setosa
14	5.8	4.0	1.2	0.2	Setosa
15	5.7	4.4	1.5	0.4	Setosa
16	5.4	3.9	1.3	0.4	Setosa
17	5.1	3.5	1.4	0.3	Setosa
18	5.7	3.8	1.7	0.3	Setosa
19	5.1	3.8	1.5	0.3	Setosa
20	5.4	3.4	1.7	0.2	Setosa
21	5.1	3.7	1.5	0.4	Setosa
22	4.6	3.6	1.0	0.2	Setosa
23	5.1	3.3	1.7	0.5	Setosa
24	4.8	3.4	1.9	0.2	Setosa
25	5.0	3.0	1.6	0.2	Setosa
26	5.0	3.4	1.6	0.4	Setosa
27	5.2	3.5	1.5	0.2	Setosa
28	5.2	3.4	1.4	0.2	Setosa
29	4.7	3.2	1.6	0.2	Setosa
30	4.8	3.1	1.6	0.2	Setosa
31	5.4	3.4	1.5	0.4	Setosa
32	5.2	4.1	1.5	0.1	Setosa
33	5.5	4.2	1.4	0.2	Setosa
34	4.9	3.1	1.5	0.2	Setosa
35	5.0	3.2	1.2	0.2	Setosa
36	5.5	3.5	1.3	0.2	Setosa
37	4.9	3.6	1.4	0.1	Setosa
38	4.4	3.0	1.3	0.2	Setosa
39	5.1	3.4	1.5	0.2	Setosa
40	5.0	3.5	1.3	0.3	Setosa
41	4.5	2.3	1.3	0.3	Setosa
42	4.4	3.2	1.3	0.2	Setosa
43	5.0	3.5	1.6	0.6	Setosa
44	5.1	3.8	1.9	0.4	Setosa
45	4.8	3.0	1.4	0.3	Setosa
46	5.1	3.8	1.6	0.2	Setosa
47	4.6	3.2	1.4	0.2	Setosa
48	5.3	3.7	1.5	0.2	Setosa
49	5.0	3.3	1.4	0.2	Setosa

In [65]:
```python
df4=df.loc[47:49]
```

In [66]:
```python
df4
```

Out[66]:

	sepal.length	sepal.width	petal.length	petal.width	variety
47	4.6	3.2	1.4	0.2	Setosa
48	5.3	3.7	1.5	0.2	Setosa
49	5.0	3.3	1.4	0.2	Setosa

In [67]:
```python
df5=df.loc[45:49]
df5
```

Out[67]:

	sepal.length	sepal.width	petal.length	petal.width	variety
45	4.8	3.0	1.4	0.3	Setosa
46	5.1	3.8	1.6	0.2	Setosa
47	4.6	3.2	1.4	0.2	Setosa
48	5.3	3.7	1.5	0.2	Setosa
49	5.0	3.3	1.4	0.2	Setosa

In [71]:
```python
df6=df.iloc[45:49,0:3]
df6
```

Out[71]:

	sepal.length	sepal.width	petal.length
45	4.8	3.0	1.4
46	5.1	3.8	1.6
47	4.6	3.2	1.4
48	5.3	3.7	1.5

In [73]:
```python
df4=df5.drop(index=[48],axis=0)
display(df4)
```

	sepal.length	sepal.width	petal.length	petal.width	variety
45	4.8	3.0	1.4	0.3	Setosa
46	5.1	3.8	1.6	0.2	Setosa
47	4.6	3.2	1.4	0.2	Setosa
49	5.0	3.3	1.4	0.2	Setosa

In [76]:
```python
df7=df4.drop('sepal.length', axis=1)
display(df7)
```

	sepal.width	petal.length	petal.width	variety
45	3.0	1.4	0.3	Setosa
46	3.8	1.6	0.2	Setosa
47	3.2	1.4	0.2	Setosa
49	3.3	1.4	0.2	Setosa

In [77]:
```python
df.columns
```

Out[77]:
```
Index(['sepal.length', 'sepal.width', 'petal.length', 'petal.width',
       'variety'],
      dtype='object')
```

In [80]:
```python
df.rename(columns={'sepal.length':'sep.len','petal.length':'pet.len'})
```

Out[80]:

	sep.len	sepal.width	pet.len	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

149 rows × 5 columns

In [81]:
```python
df.isnull()
```

Out[81]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False
149	False	False	False	False	False

149 rows × 5 columns

In [82]:
```python
df.isnull().sum()
```

Out[82]:
```
sepal.length    0
sepal.width     0
petal.length    0
petal.width     0
variety         0
dtype: int64
```

In [84]:
```
df2=df.dropna()
```

In [ ]:
```
```